Monday print day - the whole plan

Written 2026-08-01. For Mridul, printing PETG on his own Anycubic Kobra Neo.

> **The short version: you do not open the slicer at all.** The G-code is already made. Copy files to the card, press print. The only decision you make in the lab is *which folder*, and only if the test piece strings.

PART 1 - Before you leave the house

Buy these (?150 or so, any hardware/hobby shop or online)

Item	Qty	For
M2.5 brass heat-set inserts	4 + spares	The four corner posts in the cell box
M2 brass heat-set inserts	2 + spares	The two posts in the ESP32 pod

Ask for **"brass heat set threaded inserts for 3D printing"**. They look like tiny knurled brass barrels. Buy 10 of each - they cost almost nothing and you *will* lose one.

> [!] **Check the size when they arrive.** The CAD assumes the M2.5 needs a **dia 3.5mm** hole and the M2 needs **dia 3.2mm**. Brands vary. If yours are different, tell me the number and I change one line and re-slice - it takes two minutes. Do NOT print with the wrong hole size; the brass either falls straight through or splits the post.

Dry the filament

65 degC for 6 hours, in a food dehydrator or an oven on its lowest setting. Do this the night before.

PETG soaks up water from the air, and wet PETG strings no matter how good the settings are. Your June print's cobweb problem was mostly this. **No slicer setting beats dry filament.**

Copy these to the SD card

```
printing/gcode/                <- USE THIS FOLDER FIRST (230 degC)
    mid_plate.gcode
    top_plate.gcode
    outer_box.gcode
    esp32_pod_shell.gcode
    esp32_pod_lid.gcode

printing/gcode/alt_225C/       <- BACKUP, only if the first test strings
    (the same five files at 225 degC)
```

Copy **both folders**. That is the "one setting difference" you asked for - if 230 strings, you switch folders. You never touch the slicer.

PART 2 - At the printer

Print in THIS order

#	Part	Time	Why this order
1	**mid_plate**	**38 min**	[!!] **THIS IS YOUR TEST PIECE.** Smallest part, ~?15 of filament. Judge everything from it.
2	top_plate	1h 01m	Small, and it is the braille reading surface
3	outer_box	4h 29m	The big one. Only start it once the test looks right
4	esp32_pod_shell	4h 05m	Pod - not needed for the demo
5	esp32_pod_lid	1h 09m	Pod - not needed for the demo

Total ~ 11 h 20 m. If the lab closes before that, parts 1-3 are the ones that matter. **4 and 5 are the ESP32 housing and the demo does not use them.**

After part 1, stop and look at it

What you see	What it means	What to do
Clean, no cobwebs	230 degC is right	Carry on with the rest of the folder
Fine hairs between features	Too hot, or damp filament	Switch to the **alt_225C** folder
Layers splitting, part lifts off the bed	Too cold	Stay at 230, raise the **bed** to 85 degC on the printer menu
Rough, gritty top surface	Under-extruding	Tell me - it is a flow setting, not temperature

Nothing else needs judging. Those four cases cover it.

PART 3 - [!!] DO NOT PRINT `base_plate`

It is the cheapest part (44 min) **and the only one whose size depends on the cam disc.**

The resin cam arrives this week, and one hand-turn test decides whether the disc has to grow from 44mm to ~54mm. **If it grows, the base plate is the part that has to be reprinted** - its pocket holds the disc.

Every other part is independent of the disc, which is why they are safe to print now.

Holding back 44 minutes to avoid a possible reprint of 44 minutes is a free bet. Print it after the disc test.

PART 4 - Fitting the brass inserts (after printing)

You need: soldering iron, the inserts, a steady hand. **10 minutes for all six.**

```
    iron tip
       |
       v
    ___|___
   |  ___  |     1. sit the insert on the hole, knurled end down
   | |   | |     2. press the hot iron onto its top
   |_|___|_|     3. it sinks under its OWN weight - do not push
  /         \    4. stop FLUSH with the surface
 /  printed   \  5. let it cool ~30 s before touching
/_____post_____\
```

- Iron at **~220 degC**, roughly PETG printing temperature. Hotter melts too much plastic.
- **Let gravity do it.** Pressing hard squeezes molten plastic up around the brass and the thread

ends up crooked.

- Going in **tilted** is the usual failure. Look from two sides before it cools.
- If one sinks too deep or sits crooked, heat it, pull it out with the screw, let the post cool,

and retry. The plastic re-melts fine.

Practise on a scrap first if you have one.

PART 5 - What changed in the CAD since your last print

Change	Why
Electronics pocket 16 -> 14mm	Fixes a **2mm error in the whole tower**. The mid-plate rests *on top of* its ledge, not level with it, and the old arithmetic never counted the ledge. With the motor now measured at 19.0mm, taking 2mm out here puts everything back on datum - and leaves the already-ordered resin linkages valid.
Corner posts: brass insert bore added	They used to cut their own thread in PETG. That strips after a few assemblies, and this box gets opened a lot.
Pod posts grew dia 5.0 -> dia 6.5	An M2 insert needs a dia 3.2 hole. Inside a dia 5.0 post that leaves 0.9mm of wall - the hot brass would split it. dia 6.5 gives 1.65mm.
(earlier) sacrificial bridges removed, wire guides deleted, muscle-board bosses off	Documented in `CAD_CHANGELOG_v7.5.md`

PART 6 - Still unresolved, and it affects the pod lid

The power socket. Your jack is an inline pigtail with no thread and no nut, so it cannot clamp into the lid's round hole. The lid still carries a **support cradle built from invented dimensions** - it may well not fit your part.

Two ways to play it:

- **Print the lid anyway.** It is 1h 09m and the pod is not needed for the demo. If the cradle is

wrong, reprint later.

- **Send me two caliper readings** - the barrel body's **outer diameter** and its **length** - and

I will replace the invented cradle with a clamp built to your actual part before Monday.

Sixty seconds of measuring removes the guesswork entirely.

One-line summary

Dry the filament, buy the inserts, print `mid_plate` **first, look at it, then run the rest - and leave** `base_plate` **until the resin disc has been tested.**